Exercise Sheet 1

What Is a Network?

5942UE Network Science

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Prof. Dr. Michael Granitzer Dr. Kanishka Ghosh Dastidar Dr. Jelena Mitrovic

1.A What Is a Network?

Exercise 1.A.1: Recognising Networks in the Wild

For each of the five systems below, identify **(a)** what the nodes represent, **(b)** what the edges represent, **(c)** whether the edges are directed or undirected, and **(d)** whether weights are natural for the edges and what they would encode.

System	Nodes	Edges	Directed?	Weight?
University course enrollment	?	?	?	?
Wikipedia hyperlinks	?	?	?	?
Interbank lending	?	?	?	?
Power transmission grid	?	?	?	?
Co-authorship in research	?	?	?	?

After filling in the table, pick **one** system and argue: what does the network model reveal about this system that a plain list of its elements would not?

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Exercise 1.A.2: One System, Different Network Models

Return to **university course enrollment** from Exercise 1.A.1.

1. First describe it as a **bipartite network**:

   - What are the two types of nodes?
   - What do the edges represent?

2. Now transform it into **another network model** of the same system:

   - What are the nodes?
   - What do the edges represent?

3. Compare the two models:

   - What does each model help us see?
   - What information is lost or changed in the second model?

Exercise 1.A.3: The Hidden Map of Information

Imagine a high school with 500 students. The administration wants to understand how a rumour spread so quickly through the student body. They possess a database listing all 500 students, including their age, gender, grades, and extracurricular activities.

1. Why is this list of student properties insufficient to explain how the rumour travelled? What specific information is missing?
2. Propose two different types of relationships (edges) between students that might help explain the rumour's path. How would the resulting networks differ?
3. If you were allowed to look at only *one* structural feature of the true rumour-spreading network to identify the person who started it, what would you look for and why?

1.B Why Do Networks Matter?

Exercise 1.B.1: When Structure Overrides Individual Quality

Two hypothetical messaging applications launch in the same week with identical features and code quality.

- **App A** launches exclusively to 200 students at one university. Every user's existing friends are already on the platform.
- **App B** launches publicly but has no existing user base. Features are identical.

1. Which app is more likely to survive after 6 months? Argue using network concepts from the lecture.
2. The lecture showed that "knowing every person in a city does not tell you how information travels." Explain this claim using App A as a concrete example.
3. A venture capitalist says: *"The product with better technology always wins."* Based on what you learned, when is this claim wrong, and why?

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Exercise 1.B.2: Mapping Network Science Questions

Identify the core question type and analysis method for each scenario:

Scenario	Question type	Method
Vaccinating to slow a disease	?	?
Detecting informal teams in logs	?	?
Critical router failure risks	?	?
Modeling viral tweet propagation	?	?
Research community idea exchange	?	?

1.C What Can We Analyse?

Exercise 1.C.1: Structure Shapes Diffusion

Consider the high-school friendship network from the lecture. Imagine a rumour starts spreading, and at each step every student who knows the rumour tells all their friends.

Two scenarios:

- **Scenario A:** The rumour starts with a student at the dense **centre** of the network.
- **Scenario B:** The rumour starts with a student at the **periphery** (part of a short chain hanging off the main cluster).

1. In which scenario does the rumour reach more students after **2 steps**? Explain why, without calculation.
2. After enough steps, will the rumour eventually reach the same students in both scenarios? What condition determines this?
3. What does this tell us about the importance of **who** a message reaches first, independent of the message itself?